Markscheme

Mathematics: analysis and approaches
Practice question bank

162. (a)

Consider $z = -9$.

Find the square root of z with positive imaginary part, in the form $a + bi$.

$$z = 9(\cos 180^\circ + i \sin 180^\circ)$$ *(M1)*

square roots have modulus 3 and arguments $90^\circ, 270^\circ$ *(M1)*

the root with positive imaginary part is $3i$ **A1**

[3 marks]